
Does spatial context transfer? Neighbourhood-aware cell typing in seminiferous tubules under tissue-organization and species shift

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Abstract

1 Methods for annotating spatial transcriptomics data increasingly use a spot’s tissue
2 neighbourhood as input, but they are benchmarked within one tissue and condition,
3 so it is unknown whether the spatial signal they learn holds when tissue organization
4 changes. We build a small benchmark from the public Slide-seqV2 testis pucks of
5 Chen et al. (2021): three wild-type mouse, three diabetic *ob/ob* mouse, and two
6 human pucks (252,898 beads), with NMFreg cell-type labels. Using germ-cell
7 stage annotation in seminiferous tubules as the task, we compare an expression-
8 only classifier with neighbourhood-augmented features, a GraphSAGE model,
9 and test-time smoothing of the expression-only posterior, under held-out-puck,
10 wild-type-to-*ob/ob*, and mouse-to-human transfer. Spatial context adds about one
11 macro-F1 point within a species and up to three on human pucks. The *ob/ob*
12 pucks are not a distribution shift for this task. Across species, expression-only
13 accuracy drops by ten points; neighbourhood-augmented features and test-time
14 smoothing keep their gains, whereas GraphSAGE, which matches them within a
15 species, becomes the worst model, losing five points and fifteen F1 points on round
16 spermatids. The gain from context tracks how informative the neighbourhood is
17 relative to the bead itself rather than the target tissue’s neighbourhood homogeneity.
18 Fixed spatial aggregation transfers across the shifts we tested where learned spatial
19 features do not.

20 1 Introduction

21 Most recent methods for annotating cells or beads in spatial transcriptomics use the tissue neigh-
22 bourhood of a spot as an additional input, either by concatenating a spot’s own expression with an
23 aggregate of its neighbours’ [Singhal et al., 2024, Varrone et al., 2024] or by learning a graph neural
24 network over a spatial adjacency graph [Hu et al., 2021, Dong and Zhang, 2022, Long et al., 2023].
25 Benchmarks of these methods [Yuan et al., 2024, Li et al., 2022] evaluate within one tissue and one
26 condition, so they do not tell us whether the spatial signal a model learns still holds when tissue
27 organization changes, which is the question that matters when a model trained on a reference is used
28 to annotate samples from other conditions, individuals, or species.

29 The testis is a natural place to ask it. Spermatogenesis proceeds inside seminiferous tubules along
30 a radial axis, from spermatogonia at the basement membrane through spermatocytes and round
31 spermatids to elongating spermatids at the lumen, and tubule cross-sections cycle through a fixed
32 sequence of stages [Hess and de Franca, 2008, Chakravorty et al., 2026]. The same layering exists in
33 mouse and human, but the human cycle differs in stage number and in how stages tile a cross-section.
34 Chen et al. [2021] produced Slide-seqV2 [Stickels et al., 2021] pucks of wild-type mouse testis, of
35 testis from diabetic *ob/ob* mice, in which they report disrupted spatial organization of the tubules,

and of normal human testis, each with bead-level cell-type labels from NMFreg [Rodrigues et al., 2019]. This gives three levels of distribution shift on one tissue with one platform and one labelling procedure: a held-out puck of the same condition, a condition that alters tissue organization, and a different species.

We use these data as a small benchmark and ask how much spatial context improves bead-level annotation of germ-cell stage, and whether the improvement survives each shift. We compare an expression-only classifier against neighbourhood-augmented features, a graph neural network, and test-time smoothing of the expression-only posterior over the target puck’s spatial graph. The last is trained without any spatial input and uses the target tissue’s geometry only at inference, separating the value of a spatial prior from the transferability of learned spatial features.

2 Data and task

Pucks. We use the eight processed Slide-seqV2 pucks released by Chen et al. [2021]: three wild-type (WT) mouse, three *ob/ob* diabetic (DB) mouse, and two normal human (HU) pucks, each with a bead-by-gene count matrix, bead coordinates, and a per-bead NMFreg label over nine types (spermatogonia SPG, spermatocytes SPC, round spermatids RS, elongating/elongated spermatids ES, Sertoli, myoid, Leydig, endothelial, macrophage). We keep beads with a label and at least 100 UMIs, leaving 25,377 to 39,181 beads per puck (252,898 in total; Table 2).

Cross-species gene space. All pucks are restricted to one-to-one mouse–human orthologs from MGI [Baldarelli et al., 2024] detected in every puck (13,573 genes), so a model trained on mouse can be applied to human without adaptation. Counts are library-size normalized and log-transformed [Wolf et al., 2018]. We select 2,000 highly variable genes on the WT pucks only [Stuart et al., 2019], then standardize each gene within each puck using that puck’s own mean and standard deviation. This unsupervised per-puck standardization is the only batch handling we apply.

Task. The primary task is four-class germ-cell stage annotation (SPG, SPC, RS, ES) on germ-cell beads, which are 81 to 90% of beads. This is the layer of organization that the radial tubule axis defines, and it is the reproducible part of the NMFreg annotation: a classifier trained and tested on random splits of one puck reaches about the same macro-F1 as one tested on a different WT puck (0.71 vs. 0.70), whereas the rare somatic types (endothelial, macrophage, myoid) are near chance from a single bead’s expression under any split. Six-class and nine-class variants are in the appendix.

Shift settings. Training pucks are always disjoint from the test puck. *WT*→*WT*: train on two WT pucks, test on the third (three folds). *WT*→*ob/ob*: train on all three WT pucks, test on each DB puck. *WT*→*human*: train on all three WT pucks, test on each human puck; *mouse*→*human* adds the DB pucks to training. As in-distribution references for the shifted targets, *ob/ob*→*ob/ob* and *human*→*human* train on the other pucks of the same condition. The NMFreg labels were computed from expression alone, so a gain from spatial features cannot be an artefact of how the labels were made. For each puck we also compute neighbourhood homogeneity, the mean fraction of a bead’s 15 nearest spatial neighbours that share its label; it is used only to describe the pucks, never as an input. Among germ-cell beads it is 0.53 to 0.56 on WT, 0.49 to 0.57 on *ob/ob*, and 0.38 to 0.40 on human pucks.

3 Models and evaluation protocol

All models operate on a 50-dimensional PCA embedding z_i of the standardized expression, with the PCA fit on the training pucks only and applied to the test puck. Let $\mathcal{N}_k(i)$ be the k nearest spatial neighbours of bead i within its puck, and $\bar{z}_i^{(k)} = \frac{1}{k} \sum_{j \in \mathcal{N}_k(i)} z_j$ the neighbourhood mean, computed over all beads of the puck (non-germ included), since context is label-free.

Expression only. A multinomial logistic regression (LR) on z_i , and a two-layer MLP (256 hidden units, dropout 0.2). Both use class-balanced loss weights.

Table 1: Macro-F1 for germ-cell stage under six settings (left three in-distribution, right three shifted); mean $\pm$ SD across test pucks. nbr: mean embedding of the 15 nearest spatial neighbours. α selected on the WT $\rightarrow$ WT folds. MLP rows: Table 3.

Model	WT $\rightarrow$ WT	ob/ob $\rightarrow$ ob/ob	human $\rightarrow$ human	WT $\rightarrow$ ob/ob	WT $\rightarrow$ human	mouse $\rightarrow$ human
LR (expr)	0.699 $\pm$ 0.013	0.717 $\pm$ 0.022	0.486 $\pm$ 0.008	0.713 $\pm$ 0.026	0.385 $\pm$ 0.005	0.385 $\pm$ 0.006
LR (expr+nbr)	0.711 $\pm$ 0.013	0.723 $\pm$ 0.016	0.513 $\pm$ 0.011	0.721 $\pm$ 0.023	0.398 $\pm$ 0.007	0.396 $\pm$ 0.008
LR (nbr only)	0.636 $\pm$ 0.003	0.607 $\pm$ 0.019	0.464 $\pm$ 0.006	0.609 $\pm$ 0.029	0.365 $\pm$ 0.012	0.355 $\pm$ 0.008
LR (expr) + smooth ($\alpha=0.5$)	0.712 $\pm$ 0.016	0.733 $\pm$ 0.019	0.501 $\pm$ 0.008	0.732 $\pm$ 0.021	0.401 $\pm$ 0.005	0.400 $\pm$ 0.006
GraphSAGE	0.711 $\pm$ 0.015	0.716 $\pm$ 0.023	0.505 $\pm$ 0.013	0.721 $\pm$ 0.023	0.347 $\pm$ 0.012	0.350 $\pm$ 0.012

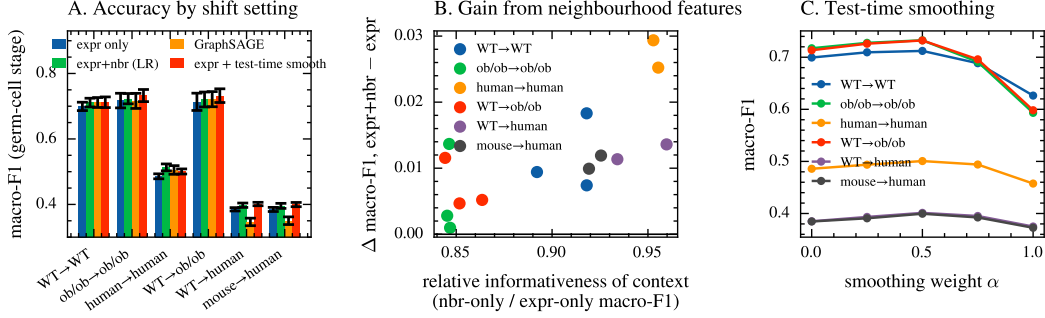


Figure 1: (A) Germ-cell stage macro-F1 by shift setting; error bars are standard deviations across test pucks. (B) Gain from neighbourhood features per test puck against the relative informativeness of context (neighbourhood-only over expression-only macro-F1). (C) Test-time smoothing of the expression-only posterior against the weight α ; $\alpha = 1$ replaces each bead’s posterior by its neighbours’ mean.

82 **Neighbourhood-augmented features.** LR and MLP on $[z_i, \bar{z}_i^{(k)}]$, the construction used by
83 BANKSY [Singhal et al., 2024] and related methods, for $k \in \{5, 15, 30\}$. As a diagnostic we
84 also fit LR on $\bar{z}_i^{(k)}$ alone, i.e. on the surroundings without the bead.

85 **Graph neural network.** A two-layer GraphSAGE with mean aggregation [Hamilton et al., 2017],
86 $h'_i = \sigma(W_1 h_i + W_2 \bar{h}_i^{(15)})$, 128 hidden units, trained full-batch on the union of training pucks (each
87 puck is its own connected graph) and applied to the test puck’s graph.

88 **Test-time smoothing.** We take the expression-only LR posterior p_i on the test puck and replace
89 it with $p'_i = (1 - \alpha) p_i + \alpha \bar{p}_i^{(15)}$, then predict $\arg \max p'_i$. This is one step of label propagation
90 on the model’s own outputs [Huang et al., 2021]; it uses the target puck’s geometry but no spatial
91 input during training, so any gain reflects the target tissue’s local homogeneity rather than transferred
92 spatial features. We sweep $\alpha \in \{0.25, 0.5, 0.75, 1\}$.

93 **Evaluation.** Macro-F1 over the classes present in the test puck and per-class F1, averaged over
94 seeds (three for MLP and GraphSAGE; LR is deterministic) and then over the test pucks of a setting,
95 with the standard deviation across test pucks. Hyperparameters were fixed a priori and shared across
96 settings (LR $C = 0.5$; AdamW with learning rate 10^{-3} for the MLP and 3×10^{-3} for GraphSAGE,
97 weight decay 10^{-4} , 40 epochs at batch size 512 and 150 full-batch epochs respectively); nothing was
98 tuned on any test puck. The full grid runs on two CPU cores in a few hours.

99 4 Results

100 **In-distribution gains are small and consistent.** On held-out WT pucks, the expression-only
101 logistic regression reaches 0.699 macro-F1, and every form of spatial context adds about one point:
102 neighbourhood-augmented LR 0.711, GraphSAGE 0.711, and test-time smoothing 0.712 (Table 1,
103 Fig. 1A). The gain is small but systematic: positive on all 15 setting–test-puck combinations for
104 both neighbourhood features and smoothing. The neighbourhood alone reaches 0.636, so most of
105 the stage signal is in the bead itself. Gains are similar on *ob/ob* and larger on human folds, where

the expression-only ceiling is lower (0.486) and neighbourhood features add 2.7 points (0.513). Neighbourhood size matters little (at most 0.008 across $k \in \{5, 15, 30\}$; Table 4). An MLP on the same features is 0.6 to 2.1 points below the LR in every setting (Table 3).

The *ob/ob* shift is small for this task. Models trained on WT pucks perform as well on *ob/ob* pucks as models trained on other *ob/ob* pucks (0.713 vs. 0.717 expression only; 0.721 vs. 0.723 with neighbourhood features). The spatial gains carry over unchanged, and germ-cell homogeneity overlaps between conditions (0.49 to 0.57 vs. 0.53 to 0.56), so at this resolution the disruption reported by Chen et al. [2021] does not register as a distribution shift for stage annotation.

Species shift is large and is dominated by expression. Expression-only macro-F1 falls from 0.486 (human→human) to 0.385 (WT→human), and adding the *ob/ob* pucks to training does not change it (0.385). Neighbourhood-augmented LR retains a gain under species shift (0.398, +1.3), as does test-time smoothing (0.401, +1.6). GraphSAGE does not: it is within 0.008 of neighbourhood-augmented LR in every in-distribution setting and becomes the worst model across species, at 0.347 (−3.8 vs. expression only, −5.1 vs. neighbourhood LR; seed-to-seed SD ≤ 0.007). Per stage (Fig. 3 in the appendix), the loss is concentrated on round spermatids, the largest human class, whose F1 drops from 0.390 to 0.237 with GraphSAGE and to 0.376 with neighbourhood LR; with neighbourhood LR, spermatogonia and spermatocytes still gain about 3 points.

What tracks the gain. The gain from context does not track the neighbourhood homogeneity of the target puck: the largest gains occur on the human pucks, which have the lowest homogeneity (0.38 to 0.40). What tracks the gain is instead how informative the neighbourhood is relative to the bead (Fig. 1B): the neighbourhood-only to expression-only ratio is 0.84 to 0.92 for mouse targets (gains of 0.1 to 1.8 points) and 0.92 to 0.96 for human targets (gains of 1.0 to 2.9). Both quantities are computed from the same labels on the same test pucks, so this is a within-sample association rather than a predictor for an unlabeled target. Test-time smoothing peaks at $\alpha = 0.5$ in every setting (Fig. 1C); full replacement ($\alpha = 1$) costs 7 to 12 points on mouse and 1 to 3 on human targets. At the bead level, the within-species gain is concentrated in the lowest-UMI tertile (+2.3 points on WT→WT, +4.7 on human→human), and context lowers accuracy for beads whose neighbourhoods are heterogeneous (Table 7).

5 Discussion

On this benchmark, spatial context is worth about one macro-F1 point within a species and up to three on human pucks, where the bead’s own expression is a weaker signal. The value is small because the neighbourhood is mostly redundant with the bead: in mouse, a bead’s stage is predictable from its 15 neighbours alone at 0.61 to 0.64 macro-F1, largely the information the bead already carries. Two further negative results follow. The *ob/ob* pucks are not a distribution shift for stage annotation, so organization changes that matter for niche-level analysis can be invisible at this level. And neighbourhood homogeneity of the target tissue does not predict how much context helps; the relative informativeness of the neighbourhood tracks it better.

The more useful result is the split between how context is used. Fixed aggregation followed by a linear model, and post-hoc smoothing of a spatial-free model, keep their gains under both shifts we tested. A two-layer GNN matches them within a species and loses about five points across species. Its first layer sees the same neighbourhood mean as the linear model; the later layers learn a nonlinear function of the mouse neighbourhood distribution that does not hold for human tubules, with the loss falling almost entirely on round spermatids. For reference-based annotation across species, a fixed spatial operator has an advantage that in-distribution benchmarks do not reveal.

Limitations. Labels are NMFreg assignments rather than ground truth, and the human labels are noisier than the mouse ones (in-distribution ceiling 0.49 vs. 0.70), so results are agreement with a published annotation, and the human estimates rest on two pucks. The cross-species gene space uses one-to-one orthologs, mouse-selected variable genes, and per-puck standardization as the only batch handling; stronger integration [Rosen et al., 2024, Lotfollahi et al., 2022] would likely narrow the species gap for all models, and whether it would close the GNN’s gap is open. We evaluated one GNN architecture at one depth, so the cross-species result is specific to this configuration. Code and splits will be released.

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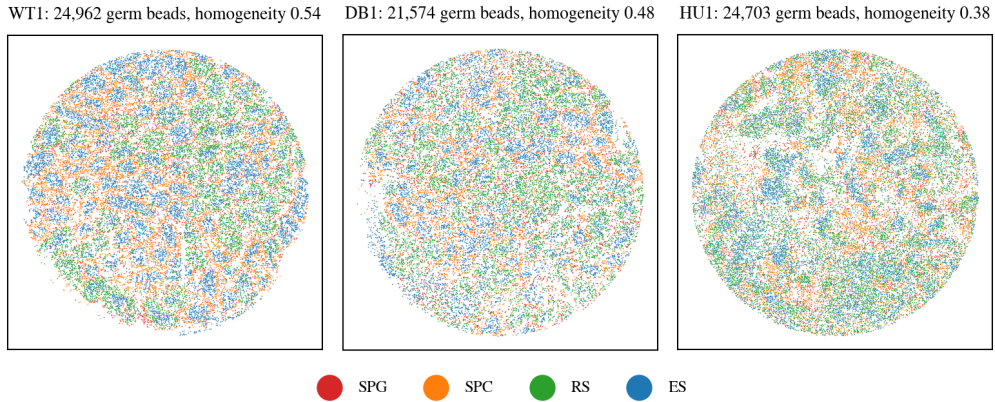
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Table 2: Puck statistics after filtering. Homogeneity is the mean fraction of a bead’s k nearest spatial neighbours that share its NMFreg label, over all beads ($k=5, 15, 30$) and over germ-cell beads only ($k=15$).

Puck	Condition	Beads	Median UMI	Homog. $k=5$	$k=15$	$k=30$	Germ, $k=15$
WT1	WT mouse	29,178	719	0.524	0.474	0.432	0.538
WT2	WT mouse	29,607	782	0.524	0.470	0.426	0.533
WT3	WT mouse	39,181	604	0.561	0.507	0.464	0.561
DB1	<i>ob/ob</i> mouse	25,377	797	0.484	0.426	0.385	0.485
DB2	<i>ob/ob</i> mouse	35,629	426	0.537	0.477	0.433	0.530
DB3	<i>ob/ob</i> mouse	30,073	684	0.574	0.518	0.473	0.568
HU1	human	30,464	602	0.362	0.336	0.316	0.380
HU2	human	33,389	600	0.379	0.352	0.331	0.397

Figure 2: Germ-cell beads of one WT mouse, one *ob/ob* mouse, and one human puck, coloured by NMFreg stage. Tubule cross-sections with a spermatogonia/spermatocyte rim and a spermatid core are visible in mouse; human tubules are larger and stage labels are more mixed.Table 3: Full version of Table 1, including the MLP models. Macro-F1, mean $\pm$ SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.699 $\pm$ 0.013	0.717 $\pm$ 0.022	0.486 $\pm$ 0.008	0.713 $\pm$ 0.026	0.385 $\pm$ 0.005	0.385 $\pm$ 0.006
LR (expr+nbr)	0.711 $\pm$ 0.013	0.723 $\pm$ 0.016	0.513 $\pm$ 0.011	0.721 $\pm$ 0.023	0.398 $\pm$ 0.007	0.396 $\pm$ 0.008
LR (nbr only)	0.636 $\pm$ 0.003	0.607 $\pm$ 0.019	0.464 $\pm$ 0.006	0.609 $\pm$ 0.029	0.365 $\pm$ 0.012	0.355 $\pm$ 0.008
LR (expr) + smooth ($\alpha=0.5$)	0.712 $\pm$ 0.016	0.733 $\pm$ 0.019	0.501 $\pm$ 0.008	0.732 $\pm$ 0.021	0.401 $\pm$ 0.005	0.400 $\pm$ 0.006
MLP (expr)	0.693 $\pm$ 0.014	0.704 $\pm$ 0.036	0.475 $\pm$ 0.012	0.704 $\pm$ 0.028	0.370 $\pm$ 0.009	0.369 $\pm$ 0.009
MLP (expr+nbr)	0.705 $\pm$ 0.018	0.708 $\pm$ 0.029	0.498 $\pm$ 0.014	0.713 $\pm$ 0.023	0.378 $\pm$ 0.013	0.375 $\pm$ 0.011
GraphSAGE	0.711 $\pm$ 0.015	0.716 $\pm$ 0.023	0.505 $\pm$ 0.013	0.721 $\pm$ 0.023	0.347 $\pm$ 0.012	0.350 $\pm$ 0.012

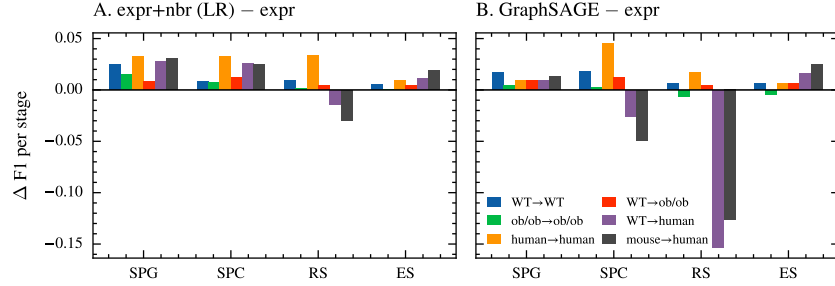


Figure 3: Per-stage change in F1 from adding spatial context, relative to the expression-only LR, for (A) neighbourhood-augmented LR and (B) GraphSAGE. Under species shift GraphSAGE loses 15 F1 points on round spermatids (RS), the most frequent human stage.

224 **Neighbourhood size and smoothing weight.** Table 4 reports the LR with neighbourhood features
 225 for $k \in \{5, 15, 30\}$ and test-time smoothing for all α .

Table 4: Sensitivity to neighbourhood size k (LR on expression plus neighbourhood mean) and to the smoothing weight α (expression-only LR posterior smoothed over the test puck’s 15-NN graph). Macro-F1 for germ-cell stage, mean over test pucks.

Setting	expr+nbr, k			smoothing weight α				
	5	15	30	0	0.25	0.5	0.75	1
WT→WT	0.715	0.711	0.708	0.699	0.709	0.712	0.689	0.626
ob/ob→ob/ob	0.728	0.723	0.720	0.717	0.728	0.733	0.691	0.593
human→human	0.511	0.513	0.510	0.486	0.494	0.501	0.494	0.457
WT→ob/ob	0.722	0.721	0.718	0.713	0.726	0.732	0.696	0.598
WT→human	0.399	0.398	0.394	0.385	0.393	0.401	0.395	0.375
mouse→human	0.397	0.396	0.390	0.385	0.391	0.400	0.392	0.373

226 **Six-class and nine-class labels.** Tables 5 and 6 repeat the linear models on all beads with a six-class
 227 scheme (four germ stages, tubular somatic, interstitial) and with the original nine NMFreg labels.
 228 Macro-F1 is much lower for nine classes because the rare somatic types (endothelial, macrophage,
 229 myoid) are near chance from a single bead’s expression even in-distribution. The gains from context
 230 are smaller than for germ-cell stage (0.4 to 1.0 points within a species for six classes, none for nine
 231 classes) but again largest on human in-distribution folds (about two points); across species with nine
 232 classes, neither neighbourhood features nor smoothing changes macro-F1 by more than 0.004.

Table 5: Six-class annotation (SPG, SPC, RS, ES, tubular somatic, interstitial), all beads. Macro-F1, mean $\pm$ SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.524 $\pm$ 0.018	0.549 $\pm$ 0.010	0.418 $\pm$ 0.001	0.547 $\pm$ 0.009	0.301 $\pm$ 0.004	0.299 $\pm$ 0.006
LR (expr+nbr)	0.533 $\pm$ 0.017	0.553 $\pm$ 0.007	0.441 $\pm$ 0.002	0.553 $\pm$ 0.007	0.308 $\pm$ 0.003	0.306 $\pm$ 0.002
LR (nbr only)	0.453 $\pm$ 0.003	0.429 $\pm$ 0.013	0.391 $\pm$ 0.006	0.435 $\pm$ 0.017	0.280 $\pm$ 0.007	0.278 $\pm$ 0.007
LR (expr) + smooth ($\alpha=0.5$)	0.531 $\pm$ 0.018	0.558 $\pm$ 0.016	0.433 $\pm$ 0.000	0.561 $\pm$ 0.012	0.316 $\pm$ 0.006	0.315 $\pm$ 0.002

Table 6: Nine-class annotation with the original NMFreg labels, all beads. Macro-F1, mean $\pm$ SD across test pucks.

Model	WT→WT	ob/ob→ob/ob	human→human	WT→ob/ob	WT→human	mouse→human
LR (expr)	0.390 $\pm$ 0.005	0.405 $\pm$ 0.000	0.309 $\pm$ 0.002	0.409 $\pm$ 0.010	0.181 $\pm$ 0.006	0.182 $\pm$ 0.003
LR (expr+nbr)	0.392 $\pm$ 0.004	0.401 $\pm$ 0.004	0.330 $\pm$ 0.001	0.413 $\pm$ 0.012	0.177 $\pm$ 0.005	0.180 $\pm$ 0.004
LR (nbr only)	0.316 $\pm$ 0.004	0.296 $\pm$ 0.010	0.280 $\pm$ 0.005	0.305 $\pm$ 0.007	0.160 $\pm$ 0.006	0.168 $\pm$ 0.002
LR (expr) + smooth ($\alpha=0.5$)	0.393 $\pm$ 0.006	0.412 $\pm$ 0.010	0.328 $\pm$ 0.001	0.416 $\pm$ 0.014	0.179 $\pm$ 0.005	0.181 $\pm$ 0.003

233 **Gain by sequencing depth and local homogeneity.** Table 7 stratifies the accuracy gain from
234 neighbourhood features by the test bead’s UMI tertile and by the fraction of its 15 neighbours
235 that share its label. Within a species the gain comes from low-UMI beads; under species shift it
236 comes from high-UMI beads. In every setting context lowers accuracy for beads with heterogeneous
237 neighbourhoods (fewer than 40% same-label neighbours) and raises it for the rest, so the puck-level
238 gain is a net of these two effects.

Table 7: Accuracy gain from neighbourhood features (expr+nbr minus expr, LR) stratified by the test bead’s UMI tertile within its puck and by its local label homogeneity. Mean over test pucks of each setting.

Setting	UMI tertile			local homogeneity				
	low	mid	high	0-0.2	0.2-0.4	0.4-0.6	0.6-0.8	0.8-1
WT→WT	+0.023	+0.003	+0.003	-0.023	-0.010	+0.022	+0.030	+0.018
ob/ob→ob/ob	+0.015	-0.001	-0.002	-0.036	-0.011	+0.017	+0.024	+0.019
human→human	+0.047	+0.021	+0.018	-0.048	+0.014	+0.091	+0.111	+0.089
WT→ob/ob	+0.015	+0.002	+0.003	-0.035	-0.009	+0.019	+0.030	+0.021
WT→human	-0.011	+0.002	+0.029	+0.000	-0.007	+0.011	+0.034	+0.050
mouse→human	-0.001	+0.004	+0.016	-0.007	-0.001	+0.013	+0.033	+0.042